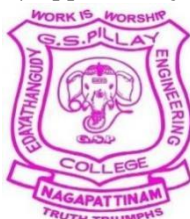


E.G.S. PILLAY ENGINEERING COLLEGE, (Autonomous)
NAGAPATTINAM – 611 002.

(Affiliated to Anna University, Chennai | Accredited by NAAC with 'A++' Grade
Accredited by NBA | Approved by AICTE, New Delhi)



REGULATIONS - R2023

B.E. / B.Tech. – SEVENTH SEMESTER CURRICULUM

Seventh Semester										
<u>S.No</u>	COURSE CODE	COURSE NAME	CATEGORY	L	T	P	C	MAX. MARKS		
								CA	ES	TOTAL
Theory Courses										
1	2302ME701	Computer Integrated Manufacturing	PCC	3	0	0	3	40	60	100
2		Professional Elective IV	PEC	3	0	0	3	40	60	100
3		Professional Elective V	PEC	3	0	0	3	40	60	100
4		Professional Elective VI	PEC	3	0	0	3	40	60	100
5		Open Elective – III	OEC	3	0	0	3	40	60	100
Laboratory Courses										
6	2302ME751	CAA and CAM Laboratory	PCC	0	0	2	1	60	40	100
7	2304ME751	Project Work - Phase -I	EEC	0	0	8	4	60	40	100
8	2304ME752	Product Development	EEC	0	0	4	2	60	40	100
			TOTAL	15	0	14	22	380	420	800

L – Lecture | T – Tutorial | P – Practical | CA – Continuous Assessment | ES – End Semester

2302ME701	COMPUTER INTEGRATED MANUFACTURING	L	T	P	C
		3	0	0	3
PREREQUISITE					
1. Manufacturing Technology I & II					
COURSE OBJECTIVES:					
To provide the overview of evolution of automation, CIM and its principles					
To learn the various Automation tools, include various material handling system.					
To train students to apply group technology and FMS.					
To familiarize the computer aided process planning in manufacturing					
To introduce to basics of data transaction, information integration and control of CIM.					
MODULE I	INTRODUCTION				9 Hours
Introduction to CAD, CAM, CAD/CAM and CIM - Evolution of CIM – CIM wheel and cycle – Production concepts and mathematical models – Simple problems in production models – CIM hardware and software – Major elements of CIM system – Three step process for implementation of CIM – Computers in CIM – Computer networks for manufacturing – The future automated factory – Management of CIM – safety aspects of CIM– advances in CIM					
MODULE II	AUTOMATED MANUFACTURING SYSTEMS				9 Hours
Automated production line – system configurations, work part transfer mechanisms – Fundamentals of Automated assembly system – System configuration, Part delivery at workstations – Design for automated assembly – Overview of material handling equipments – Consideration in material handling system design – The 10 principles of Material handling. Conveyor systems – Types of conveyors – Operations and features. Automated Guided Vehicle system – Types & applications – Vehicle guidance technology – Vehicle management and safety. Storage system performance – storage location strategies – Conventional storage methods and equipments – Automated storage/Retrieval system and Carousel storage system Deadlocks in Automated manufacturing systems – Petrinet models – Applications in Dead lock avoidance – smart manufacturing – Industry 4.0 - Digital manufacturing – Virtual manufacturing					
MODULE III	GROUP TECHNOLOGY AND FMS				9 Hours
Part families – Visual – Parts classification and coding – Production flow analysis – Grouping of parts and Machines by rank order clustering method – Benefits of GT – Case studies. FMS – Components – workstations – FMS layout configurations – Computer control systems – FMS planning and implementation issues – Architecture of FMS – flow chart showing various operations in FMS – Machine cell design – Composite part concept, Holier method, Key machine concept – Quantitative analysis of FMS – Bottleneck model – Simple and complicated problems – Extended Bottleneck model - sizing the FMS – FMS applications, Benefits.					
MODULE IV	PROCESS PLANNING				9 Hours
Process planning – Activities in process planning, Informations required. From design to process planning – classification of manufacturing processes – Selection of primary manufacturing processes – Sequencing of operations according to Anteriorities – various examples – forming of Matrix of Anteriorities – case study. Typical process sheet – case studies in Manual process planning. Computer Aided Process Planning – Process planning module and data base – Variant process planning – Two stages in VPP – Generative process planning – Flow chart showing various activities in generative PP – Semi generative process planning- Comparison of CAPP and Manual PP.					
MODULE V	PROCESS CONTROL AND DATA ANALYSIS				9 Hours
Introduction to process model formulation – linear feedback control systems – Optimal control – Adaptive control –Sequence control and PLC& SCADA. Computer process control – Computer process interface – Interface hardware – Computer process monitoring – Direct digital control and Supervisory computer control - Overview of Automatic identification methods – Bar code technology –Automatic data capture technologies - Quality management (SPC) and automated inspection					

TOTAL: 45 HOURS												
COURSE OUTCOMES:												
On the successful completion of the course, students will be able to												
CO1:	Discuss the basics of computer aided engineering											
CO2:	Choose appropriate automotive tools and material handling systems											
CO3:	Discuss the overview of group technology, FMS and automation identification methods.											
CO4:	Design using computer aided process planning for manufacturing of various components											
CO5:	Acquire knowledge in computer process control techniques.											
COs Vs POs MAPPING:												
Cos	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	1	2				1		1	
CO2	3	2	2	1	2				1		1	
CO3	3	2	2	1	2				1		1	
CO4	3	2	2	1	2				1		1	
CO5	3	2	2	1	2				1		1	
COs Vs PSOs MAPPING:												
				COs	PSO1	PSO2	PSO3					
				CO1		2						
				CO2		2						
				CO3		2						
				CO4		2						
				CO5		2						
REFERENCES:												
1. Shivanand H K, Benal M M and Koti V, Flexible Manufacturing System, New Age, 2016												
2. CIM: Computer Integrated Manufacturing: Computer Steered Industry Book by August-Wilhelm Schee												
3. 1Alavudeen and Venkateshwaran, Computer Integrated Manufacturingl, PHI Learning Pvt. Ltd., New Delhi, 2013.												
4. James A. Retr, Herry W. Kraebber, Computer Integrated Manufacturingl, Pearson Education, Asia,3rdEdition,2004.												
5. Mikell P. Groover, Automation, Production system and Computer integrated Manufacturing, Prentice Hall of India Pvt. Ltd., 4thEdition, 2014.												
6. 2Radhakrishnan P, Subramanian S and Raju V, CAD/CAM/CIM, New Age International Publishers, 3rd Edition, 2008.												

2302ME751	CAA AND CAM LABORATORY				L	T	P	C			
					0	0	2	1			
PREREQUISITE											
1. Computer Aided Design 2. Strength of Materials 3. Manufacturing Processes											
COURSE OBJECTIVES											
1. To develop hands-on proficiency in 3D modeling 2. To provide exposure to Computer Aided Manufacturing (CAM) 3. To impart knowledge of industry-standard software tools 4. To develop an understanding of CNC machine operation and programming 5. To bridge the gap between design and manufacturing											
List of Experiments											
Computer Aided Analysis											
1. Stress and deflection analysis in beams with different support conditions. 2. Stress analysis of bracket. 3. Thermal stress analysis of mixed boundary. 4. Model analysis of Beams. 5. Harmonic analysis of simple systems. 6. Stress analysis of 3D beam.											
Computer Aided Manufacturing											
1. Part programming for a Pocketing operation using CNC Milling Software. 2. Part programming for a Contouring operation using CNC Milling Software. 3. Part programming for a Drilling operation using CNC Milling Software. 4. Part programming for a Turning operation using CNC Lathe Software. 5. Part programming for a Taper turning operation using CNC Lathe Software. 6. Part programming for a Undercut operation using CNC Lathe Software.											
TOTAL: 30 HOURS											
COURSE OUTCOMES:											
After completing this course, students will be able to:											
CO1:	Create 3D models of mechanical components and assemblies using CAD software tools.										
CO2:	Perform basic Finite Element Analysis (FEA) to evaluate stresses, strains, and deformation in mechanical components.										
CO3:	Generate and verify tool paths for CNC machining using CAM software.										
CO4:	Apply G-code and M-code for CNC machine programming and operation.										
CO5:	Analyze and optimize component designs based on simulation feedback for improved manufacturability and performance.										
COs Vs POs MAPPING:											
Cos	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	2	2	3	2	3	-	-	-	2	-	2
CO2	3	2	3	2	3	-	-	-	2	-	2
CO3	3	2	3	2	3	-	-	-	2	-	2
CO4	2	1	3	2	3	-	-	-	2	-	2
CO5	3	2	3	2	3	-	-	-	2	-	2
COs Vs PSOs MAPPING:											

	CO1	-	-	3	
	CO2	-	-	3	
	CO3	-	2	3	
	CO4	-	2	3	
	CO5	-	2	3	

References:

1. Ibrahim Zeid, CAD/CAM Theory and Practice, McGraw-Hill Education.
2. Mikell P. Groover, Automation, Production Systems, and Computer-Integrated Manufacturing, Pearson.
3. P.N. Rao, CAD/CAM: Principles and Applications, Tata McGraw-Hill.
4. S. K. Sinha, CNC Programming using Fanuc CNC System, Galgotia Publications.
5. David F. Rogers & J. Alan Adams, Mathematical Elements for Computer Graphics, McGraw-Hill

2304ME752	PRODUCT DEVELOPMENT					L	T	P	C			
						0	0	4	2			
PREREQUISITE												
1. All Courses												
COURSE OBJECTIVES												
- To encourage students to identify, design, and fabricate components or systems based on real-life engineering problems. - To develop the ability to apply theoretical knowledge in practical engineering design and manufacturing. - To improve team collaboration, project planning, documentation, and communication skills. - To expose students to hands-on experience in using machine tools, fabrication techniques, and design software. - To inculcate problem-solving, innovation, and critical thinking in the development of mechanical systems.												
GUIDELINE FOR REVIEW AND EVALUATION												
The students in a group of 3 to 4 works on a topic approved by the head of the department under the guidance of a faculty member and prepares a comprehensive project report after completing the work to the satisfaction of the supervisor. The progress of the project is evaluated based on a minimum of three reviews. The review committee may be constituted by the Head of the Department. A project report is required at the end of the semester. The project work is evaluated based on oral presentation and the project report jointly by external and internal examiners constituted by the Head of the Department.												
TOTAL: 30 HOURS												
COURSE OUTCOMES:												
On the successful completion of the course, students will be able to												
CO1:	Identify and define engineering problems suitable for mini project development.											
CO2:	Develop conceptual design and perform analysis using engineering principles											
CO3:	Select appropriate materials and manufacturing processes for fabrication.											
CO4:	Fabricate mechanical components or systems using machine tools and techniques.											
CO5:	Demonstrate teamwork, documentation, and presentation skills effectively.											
COs Vs POs MAPPING:												
	Cos	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
	CO1	3	2	2	1	1				2	1	
	CO2	3	3	3	2	2				1	1	1
	CO3	2	2	3	1	2						1
	CO4	2	2	3	2	3						
	CO5	1	1	2	1	1			1	3	3	1
COs Vs PSOs MAPPING:												
		Cos	PSO1	PSO2	PSO3							
		CO1		3	3							
		CO2		3	3							
		CO3		3	3							
		CO4		3	3							
		CO5		3	3							